

Machine Learning II - Structured Models

Exercise sheet **Duality in Convex Programming. Quadratic Programs**

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Construct the Lagrange dual for the problem:

Exercise 1. Compute the dual problem. Compare primal and dual optimal values. Can you compute primal solution from the dual one?

$$\min \frac{1}{2}(x^2 + y^2) \quad (1)$$

$$x \leq y \quad (2)$$

$$-x \leq y \quad (3)$$

Exercise 2.

$$\min (a - x)^2 + c \quad (4)$$

$$Ax = b \quad (5)$$

$$x \geq 0 \quad (6)$$

Check that only trivial inequalities appear in the dual problem.

Exercise 3.

$$\min_{x,y} (a - y)^2 + bx + c \quad (7)$$

$$Ax = b \quad (8)$$

$$x \geq 0 \quad (9)$$

Primal objective is linear w.r.t. x . This leads to non-trivial linear inequalities in the dual.

Exercise 4 (Lagrange Dual of QP).

1. "Separable linear discriminative learning":

$$\min_w \frac{C}{2} \|w\|_2^2 \quad (10)$$

$$\langle w, x^i \rangle \geq 1, \quad i = 1, \dots, m \quad (11)$$

Show that its dual is also QP and its objective can be represented via a kernel matrix κ , where $\kappa_{ij} = \langle x^i, x^j \rangle$.

2. "Binary SVM": $\mathcal{X} = \mathcal{X}^0 \cup \mathcal{X}^1$ - finite sets

$$\min_{w,\xi} \frac{1}{2} \|w\|_2^2 + C \sum_{i=1}^n \xi_i \quad (12)$$

$$y_i(\langle w, x^i \rangle - b) \geq 1 - \xi_i, \quad i = 1, \dots, n, \quad y_i = 1 \text{ if } x^i \in \mathcal{X}^0 \text{ and } y_i = -1 \text{ if } x^i \in \mathcal{X}^1 \quad (13)$$

$$\xi_i \geq 0 \quad (14)$$

Show that its dual is also QP and its objective can be represented via a kernel matrix κ , where $\kappa_{ij} = \langle x^i, x^j \rangle$.

Fill out the Table with your results.